

Assignment -02

SQL-based data analysis using filtering, aggregation, and basic business queries.

1. Load dataset into a SQL database:

- Firstly create a database superstore_db and use it.
- Then create a table named superstore matching the given dataset columns.
- Import CSV file from left navigation panel SCEMAS.
- Right-click superstore table from superstore_db database and select Table data Import Wizard.

```
CREATE DATABASE superstore_db;
```

```
USE superstore_db;
```

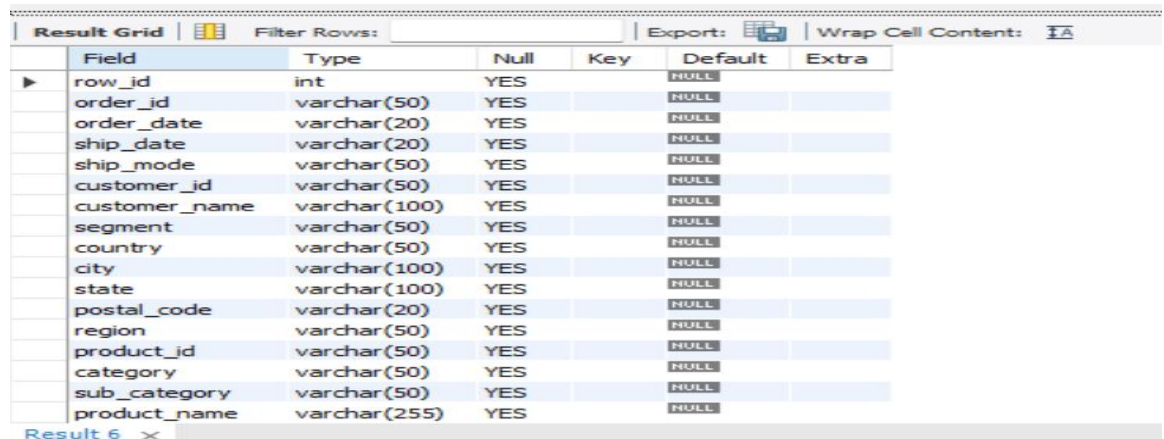
```
CREATE TABLE superstore (  
    row_id INT,  
    order_id VARCHAR(50),  
    order_date VARCHAR(20),  
    ship_date VARCHAR(20),  
    ship_mode VARCHAR(50),  
    customer_id VARCHAR(50),  
    customer_name VARCHAR(100),  
    segment VARCHAR(50),  
    country VARCHAR(50),  
    city VARCHAR(100),  
    state VARCHAR(100),  
    postal_code VARCHAR(20),  
    region VARCHAR(50),  
    product_id VARCHAR(50),  
    category VARCHAR(50),  
    sub_category VARCHAR(50),  
    product_name VARCHAR(255),  
    sales DECIMAL(10,2),  
    quantity INT,  
    discount DECIMAL(5,2),  
    profit DECIMAL(10,2)
```

);

2. Explore table (schema, sample data).

- View Table structure:

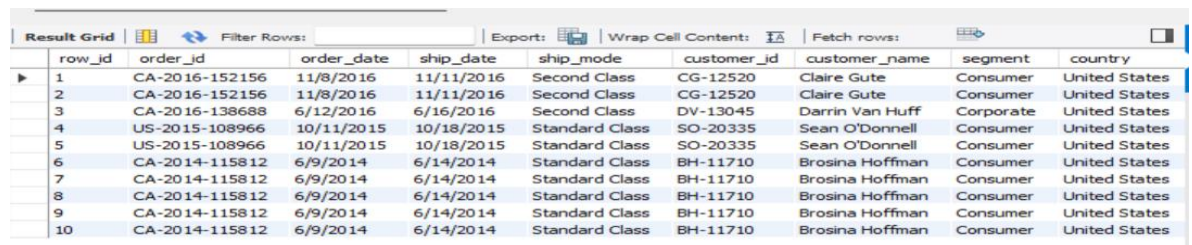
SHOW COLUMNS FROM superstore;



Field	Type	Null	Key	Default	Extra
row_id	int	YES		NULL	
order_id	varchar(50)	YES		NULL	
order_date	varchar(20)	YES		NULL	
ship_date	varchar(20)	YES		NULL	
ship_mode	varchar(50)	YES		NULL	
customer_id	varchar(50)	YES		NULL	
customer_name	varchar(100)	YES		NULL	
segment	varchar(50)	YES		NULL	
country	varchar(50)	YES		NULL	
city	varchar(100)	YES		NULL	
state	varchar(100)	YES		NULL	
postal_code	varchar(20)	YES		NULL	
region	varchar(50)	YES		NULL	
product_id	varchar(50)	YES		NULL	
category	varchar(50)	YES		NULL	
sub_category	varchar(50)	YES		NULL	
product_name	varchar(255)	YES		NULL	

- View Sample data:

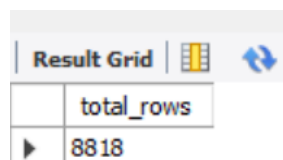
SELECT *FROM superstore LIMIT 10;



row_id	order_id	order_date	ship_date	ship_mode	customer_id	customer_name	segment	country
1	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States
2	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States
3	CA-2016-138688	6/12/2016	6/16/2016	Second Class	DV-13045	Darrin Van Huff	Corporate	United States
4	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States
5	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States
6	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States
7	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States
8	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States
9	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States
10	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States

- Count total records;

SELECT COUNT(*) AS total_rows FROM superstore;



total_rows
8818

3. Apply WHERE filters (region, category, date, sales)

- Filter by region:
 - Showing all records from west region

SELECT *FROM superstore WHERE Region = 'West';

row_id	order_id	order_date	ship_date	ship_mode	customer_id	customer_name	segment	country
3	CA-2016-136688	6/12/2016	6/16/2016	Second Class	DV-13045	Darrin Van Huff	Corporate	United State
6	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United State
7	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United State
8	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United State
9	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United State
10	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United State
11	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United State
12	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United State
14	CA-2016-161389	12/5/2016	12/10/2016	Standard Class	IM-15070	Irene Maddox	Consumer	United State
18	CA-2014-167164	5/13/2014	5/15/2014	Second Class	AG-10270	Alejandro Grove	Consumer	United State
19	CA-2014-143336	8/27/2014	9/1/2014	Second Class	ZD-21925	Zuschuss Donatelli	Consumer	United State
20	CA-2014-143336	8/27/2014	9/1/2014	Second Class	ZD-21925	Zuschuss Donatelli	Consumer	United State
21	CA-2014-143336	8/27/2014	9/1/2014	Second Class	ZD-21925	Zuschuss Donatelli	Consumer	United State
25	CA-2015-106320	9/25/2015	9/30/2015	Standard Class	EB-13870	Emily Burns	Consumer	United State
26	CA-2016-121755	1/16/2016	1/20/2016	Second Class	EH-13945	Eric Hoffmann	Consumer	United State
27	CA-2016-121755	1/16/2016	1/20/2016	Second Class	EH-13945	Eric Hoffmann	Consumer	United State

- Showing all records from east region

SELECT *FROM superstore WHERE Region = 'East';

row_id	order_id	order_date	ship_date	ship_mode	customer_id	customer_name	segment	country
24	US-2017-156909	7/16/2017	7/18/2017	Second Class	SF-20065	Sandra Flanagan	Consumer	United State
28	US-2015-150630	9/17/2015	9/21/2015	Standard Class	TB-21520	Tracy Blumstein	Consumer	United State
29	US-2015-150630	9/17/2015	9/21/2015	Standard Class	TB-21520	Tracy Blumstein	Consumer	United State
30	US-2015-150630	9/17/2015	9/21/2015	Standard Class	TB-21520	Tracy Blumstein	Consumer	United State
31	US-2015-150630	9/17/2015	9/21/2015	Standard Class	TB-21520	Tracy Blumstein	Consumer	United State
32	US-2015-150630	9/17/2015	9/21/2015	Standard Class	TB-21520	Tracy Blumstein	Consumer	United State
34	US-2015-150630	9/17/2015	9/21/2015	Standard Class	TB-21520	Tracy Blumstein	Consumer	United State
48	CA-2016-169194	6/20/2016	6/25/2016	Standard Class	LH-16900	Lena Hernandez	Consumer	United State
49	CA-2016-169194	6/20/2016	6/25/2016	Standard Class	LH-16900	Lena Hernandez	Consumer	United State
54	CA-2016-105816	12/11/2016	12/17/2016	Standard Class	JM-15265	Janet Molinari	Corporate	United State
55	CA-2016-105816	12/11/2016	12/17/2016	Standard Class	JM-15265	Janet Molinari	Corporate	United State
56	CA-2016-111682	6/17/2016	6/18/2016	First Class	TB-21055	Ted Butterfield	Consumer	United State
57	CA-2016-111682	6/17/2016	6/18/2016	First Class	TB-21055	Ted Butterfield	Consumer	United State
58	CA-2016-111682	6/17/2016	6/18/2016	First Class	TB-21055	Ted Butterfield	Consumer	United State
59	CA-2016-111682	6/17/2016	6/18/2016	First Class	TB-21055	Ted Butterfield	Consumer	United State
60	CA-2016-111682	6/17/2016	6/18/2016	First Class	TB-21055	Ted Butterfield	Consumer	United State

- Filter by Category
- Showing all furniture products

SELECT *
FROM superstore
WHERE Category = 'Furniture';

row_id	order_id	order_date	ship_date	ship_mode	customer_id	customer_name	segment	country
1	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United State
2	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United State
4	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United State
6	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United State
11	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United State
24	US-2017-156909	7/16/2017	7/18/2017	Second Class	SF-20065	Sandra Flanagan	Consumer	United State
25	CA-2015-106320	9/25/2015	9/30/2015	Standard Class	EB-13870	Emily Burns	Consumer	United State
28	US-2015-150630	9/17/2015	9/21/2015	Standard Class	TB-21520	Tracy Blumstein	Consumer	United State
30	US-2015-150630	9/17/2015	9/21/2015	Standard Class	TB-21520	Tracy Blumstein	Consumer	United State
37	CA-2016-117590	12/8/2016	12/10/2016	First Class	GH-14485	Gene Hale	Corporate	United State
39	CA-2015-117415	12/27/2015	12/31/2015	Standard Class	SN-20710	Steve Nguyen	Home Off...	United State
40	CA-2015-117415	12/27/2015	12/31/2015	Standard Class	SN-20710	Steve Nguyen	Home Off...	United State
52	CA-2015-115742	4/18/2015	4/22/2015	Standard Class	DP-13000	Darren Powers	Consumer	United State
53	CA-2015-115742	4/18/2015	4/22/2015	Standard Class	DP-13000	Darren Powers	Consumer	United State
58	CA-2016-111682	6/17/2016	6/18/2016	First Class	TB-21055	Ted Butterfield	Consumer	United State
66	CA-2015-135545	11/24/2015	11/30/2015	Standard Class	KM-16720	Kunst Miller	Consumer	United State

- Showing all technology products

SELECT *
FROM superstore
WHERE Category = 'Technology';

Result Grid									
Filter Rows:									
Export: Wrap Cell Content: Fetch rows:									
row_id	order_id	order_date	ship_date	ship_mode	customer_id	customer_name	segment	country	
8	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United S	
12	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United S	
20	CA-2014-143336	8/27/2014	9/1/2014	Second Class	ZD-21925	Zuschuss Donatelli	Consumer	United S	
27	CA-2016-121755	1/16/2016	1/20/2016	Second Class	EH-13945	Eric Hoffmann	Consumer	United S	
36	CA-2016-117590	12/8/2016	12/10/2016	First Class	GH-14485	Gene Hale	Corporate	United S	
41	CA-2015-117415	12/27/2015	12/31/2015	Standard Class	SN-20710	Steve Nguyen	Home Office	United S	
42	CA-2017-120999	9/10/2017	9/15/2017	Standard Class	LC-16930	Linda Cazamias	Corporate	United S	
45	CA-2016-118255	3/11/2016	3/13/2016	First Class	ON-18715	Odella Nelson	Corporate	United S	
48	CA-2016-169194	6/20/2016	6/25/2016	Standard Class	LH-16900	Lena Hernandez	Consumer	United S	
49	CA-2016-169194	6/20/2016	6/25/2016	Standard Class	LH-16900	Lena Hernandez	Consumer	United S	
55	CA-2016-105816	12/11/2016	12/17/2016	Standard Class	JM-15265	Janet Molinari	Corporate	United S	
60	CA-2016-111682	6/17/2016	6/18/2016	First Class	TB-21055	Ted Butterfield	Consumer	United S	
63	CA-2015-135545	11/24/2015	11/30/2015	Standard Class	KM-16720	Kunst Miller	Consumer	United S	
69	CA-2014-106376	12/5/2014	12/10/2014	Standard Class	BS-11590	Brendan Sweed	Corporate	United S	
87	CA-2017-155558	10/26/2017	11/2/2017	Standard Class	PG-18895	Paul Gonzalez	Consumer	United S	
91	CA-2016-109806	9/17/2016	9/22/2016	Standard Class	JS-15685	Jim Sink	Corporate	United S	

- Filter by Sales
- Sales greater than 1000

```
SELECT *
FROM superstore
WHERE Sales > 1000;
```

Result Grid									
Filter Rows:									
Export: Wrap Cell Content:									
row_id	order_id	order_date	ship_date	ship_mode	customer_id	customer_name	segment	country	
11	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United State	
25	CA-2015-106320	9/25/2015	9/30/2015	Standard Class	EB-13870	Emily Burns	Consumer	United State	
28	US-2015-150630	9/17/2015	9/21/2015	Standard Class	TB-21520	Tracy Blumstein	Consumer	United State	
36	CA-2016-117590	12/8/2016	12/10/2016	First Class	GH-14485	Gene Hale	Corporate	United State	
55	CA-2016-105816	12/11/2016	12/17/2016	Standard Class	JM-15265	Janet Molinari	Corporate	United State	
68	CA-2014-106376	12/5/2014	12/10/2014	Standard Class	BS-11590	Brendan Sweed	Corporate	United State	
150	CA-2016-114489	12/5/2016	12/9/2016	Standard Class	JE-16165	Justin Ellison	Corporate	United State	
166	CA-2014-139892	9/8/2014	9/12/2014	Standard Class	BM-11140	Becky Martin	Consumer	United State	
168	CA-2014-139892	9/8/2014	9/12/2014	Standard Class	BM-11140	Becky Martin	Consumer	United State	

- Sales between 500 and 2000
- ```
SELECT *
FROM superstore
WHERE Sales BETWEEN 500 AND 2000;
```

| Result Grid                |                |            |            |                |             |                 |             |            |  |
|----------------------------|----------------|------------|------------|----------------|-------------|-----------------|-------------|------------|--|
| Filter Rows:               |                |            |            |                |             |                 |             |            |  |
| Export: Wrap Cell Content: |                |            |            |                |             |                 |             |            |  |
| row_id                     | order_id       | order_date | ship_date  | ship_mode      | customer_id | customer_name   | segment     | country    |  |
| 2                          | CA-2016-152156 | 11/8/2016  | 11/11/2016 | Second Class   | CG-12520    | Claire Gute     | Consumer    | United Stz |  |
| 4                          | US-2015-108966 | 10/11/2015 | 10/18/2015 | Standard Class | SO-20335    | Sean O'Donnell  | Consumer    | United Stz |  |
| 8                          | CA-2014-115812 | 6/9/2014   | 6/14/2014  | Standard Class | BH-11710    | Brosina Hoffman | Consumer    | United Stz |  |
| 11                         | CA-2014-115812 | 6/9/2014   | 6/14/2014  | Standard Class | BH-11710    | Brosina Hoffman | Consumer    | United Stz |  |
| 12                         | CA-2014-115812 | 6/9/2014   | 6/14/2014  | Standard Class | BH-11710    | Brosina Hoffman | Consumer    | United Stz |  |
| 17                         | CA-2014-105893 | 11/11/2014 | 11/18/2014 | Standard Class | PK-19075    | Pete Kriz       | Consumer    | United Stz |  |
| 25                         | CA-2015-106320 | 9/25/2015  | 9/30/2015  | Standard Class | EB-13870    | Emily Burns     | Consumer    | United Stz |  |
| 36                         | CA-2016-117590 | 12/8/2016  | 12/10/2016 | First Class    | GH-14485    | Gene Hale       | Corporate   | United Stz |  |
| 39                         | CA-2015-117415 | 12/27/2015 | 12/31/2015 | Standard Class | SN-20710    | Steve Nauven    | Home Office | United Stz |  |

- Filter by Dates:
- ```
SELECT *
FROM superstore
WHERE YEAR(STR_TO_DATE(order_date,'%m/%d/%Y')) = 2017;
```

Result Grid									
Filter Rows:									
Export: Wrap Cell Content: Fetch rows:									
row_id	order_id	order_date	ship_date	ship_mode	customer_id	customer_name	segment	country	
13	CA-2017-114412	4/15/2017	4/20/2017	Standard Class	AA-10480	Andrew Allen	Consumer	United S	
24	US-2017-156909	7/16/2017	7/18/2017	Second Class	SF-20065	Sandra Flanagan	Consumer	United S	
35	CA-2017-107727	10/19/2017	10/23/2017	Second Class	MA-17560	Matt Abelman	Home Office	United S	
42	CA-2017-120999	9/10/2017	9/15/2017	Standard Class	LC-16930	Linda Cazamias	Corporate	United S	
44	CA-2017-139619	9/19/2017	9/23/2017	Standard Class	ES-14080	Erin Smith	Corporate	United S	
72	CA-2017-114440	9/14/2017	9/17/2017	Second Class	TB-21520	Tracy Blumstein	Consumer	United S	
76	US-2017-118038	12/9/2017	12/11/2017	First Class	KB-16600	Ken Brennan	Corporate	United S	
78	US-2017-118038	12/9/2017	12/11/2017	First Class	KB-16600	Ken Brennan	Corporate	United S	
85	US-2017-110667	11/13/2017	11/16/2017	First Class	CS-12400	Christopher Schild	Home Office	United S	

- Multiple Conditions:

Ex: Technology products in West region

```
SELECT *
FROM superstore
WHERE Region = 'West'
AND Category = 'Technology';
```

Result Grid									
Filter Rows:									
Export: Wrap Cell Content:									
row_id	order_id	order_date	ship_date	ship_mode	customer_id	customer_name	segment	country	
8	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United S	
12	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United S	
20	CA-2014-143336	8/27/2014	9/1/2014	Second Class	ZD-21925	Zuschuss Donatelli	Consumer	United S	
27	CA-2016-121755	1/16/2016	1/20/2016	Second Class	EH-13945	Eric Hoffmann	Consumer	United S	
63	CA-2015-135545	11/24/2015	11/30/2015	Standard Class	KM-16720	Kunst Miller	Consumer	United S	
69	CA-2014-106376	12/5/2014	12/10/2014	Standard Class	BS-11590	Brendan Sweed	Corporate	United S	
91	CA-2016-109806	9/17/2016	9/22/2016	Standard Class	JS-15685	Jim Sink	Corporate	United S	
104	US-2015-156867	11/13/2015	11/17/2015	Standard Class	LC-16870	Lena Cacioppo	Consumer	United S	

Brief Insights:

- The West region contains a significant number of orders.
- Technology products have several high-value transactions.
- Orders with sales above \$1000 represent premium purchases.

4. Use GROUP BY for aggregations (sales, quantity, averages).

- Total Sales by Region
SELECT Region,
SUM(Sales) AS Total_Sales
FROM superstore
GROUP BY Region;

Result Grid			Filter Rows
	Region	Total_Sales	
▶	South	369086.20	
	West	645382.70	
	Central	458653.75	
	East	604481.92	

2. Total Sales by Category

```
SELECT Category,
       SUM(Sales) AS Total_Sales
FROM superstore
GROUP BY Category;
```

Result Grid			Filter Rows:
	Category	Total_Sales	
▶	Furniture	659062.14	
	Office Supplies	636828.79	
	Technology	781713.64	

3. Total Quantity Sold by Category

```
SELECT Category,
       SUM(Quantity) AS Total_Quantity
FROM superstore
GROUP BY Category;
```

Result Grid			Filter Rows:
	Category	Total_Quantity	
▶	Furniture	7186	
	Office Supplies	19942	
	Technology	6308	

4. Average Sales by Category

```
SELECT Category,
       AVG(Sales) AS Average_Sales
FROM superstore
GROUP BY Category;
```

Result Grid   Filter Rows:		
	Category	Average_Sales
▶	Furniture	348.710127
	Office Supplies	121.370076
	Technology	465.028935

5. Average Profit by Region

```
SELECT Region,
       AVG(Profit) AS Average_Profit
FROM superstore
GROUP BY Region;
```

Result Grid   Filter Rows:		
	Region	Average_Profit
▶	South	29.919027
	West	34.175030
	Central	20.213688
	East	32.826275

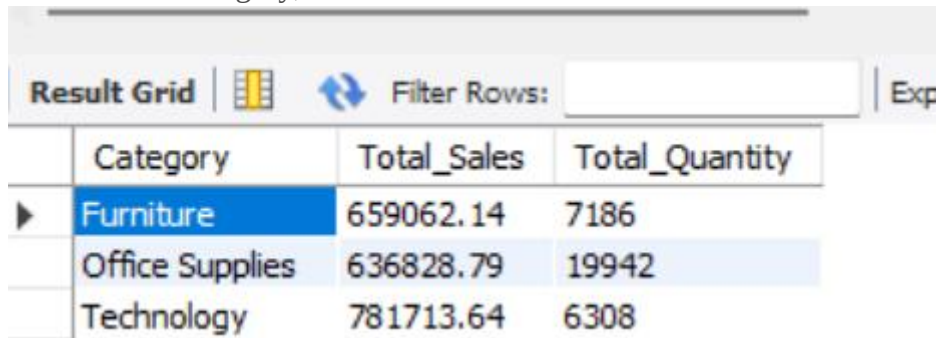
6. Total Profit by Category

```
SELECT Category,
       SUM(Profit) AS Total_Profit
FROM superstore
GROUP BY Category;
```

Result Grid   Filter Rows:		
	Category	Total_Profit
▶	Furniture	15248.27
	Office Supplies	111087.76
	Technology	136900.32

7. Sales and Quantity Together

```
SELECT Category,  
       SUM(Sales) AS Total_Sales,  
       SUM(Quantity) AS Total_Quantity  
FROM superstore  
GROUP BY Category;
```



The screenshot shows a 'Result Grid' interface with a table of query results. The table has four columns: 'Category', 'Total_Sales', and 'Total_Quantity'. There are three rows of data. The first row is 'Furniture' with 'Total_Sales' of 659062.14 and 'Total_Quantity' of 7186. The second row is 'Office Supplies' with 'Total_Sales' of 636828.79 and 'Total_Quantity' of 19942. The third row is 'Technology' with 'Total_Sales' of 781713.64 and 'Total_Quantity' of 6308. The 'Furniture' row is highlighted in blue.

	Category	Total_Sales	Total_Quantity
▶	Furniture	659062.14	7186
	Office Supplies	636828.79	19942
	Technology	781713.64	6308

Brief Insights:

- Technology generated the highest total sales.
- Office Supplies sold the highest quantity of products.
- The West region contributed the most revenue.
- Some categories have high sales but comparatively lower profits.
- Customer purchasing behavior varies significantly across regions.

5. Sort and limit results (top products, top categories).

1. Top 10 Products by Sales

```
SELECT product_name,  
       SUM(Sales) AS Total_Sales  
FROM superstore  
GROUP BY product_name  
ORDER BY Total_Sales DESC  
LIMIT 10;
```


Result Grid   Filter Rows: Export:  W		
product_name	Total_Sales	
Cisco TelePresence System EX90 Videoconferen...	22638.48	
GBC Ibimaster 500 Manual ProClick Binding System	19024.50	
Hewlett Packard LaserJet 3310 Copier	18839.68	
HP Designjet T520 Inkjet Large Format Printer -...	18374.90	
High Speed Automatic Electric Letter Opener	17030.31	
Lexmark MX611dhe Monochrome Laser Printer	16829.90	
Martin Yale Chadless Opener Electric Letter Ope...	16656.21	
Ibico EPK-21 Electric Binding System	15875.92	

2. Top Categories by Sales

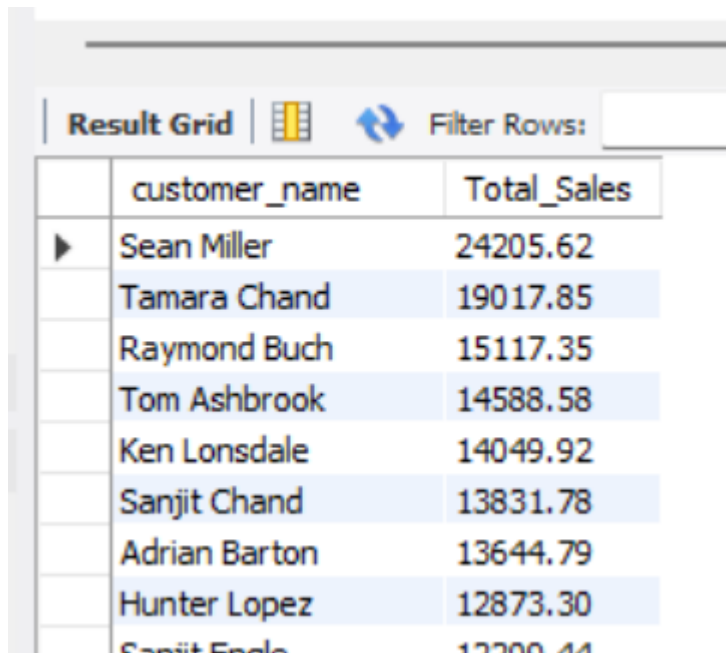
```
SELECT Category,
       SUM(Sales) AS Total_Sales
FROM superstore
GROUP BY Category
ORDER BY Total_Sales DESC;
```

Result Grid   Filter Rows:		
	Category	Total_Sales
▶	Technology	781713.64
	Furniture	659062.14
	Office Supplies	636828.79

3. Top 10 Customers by Sales

```
SELECT `Customer Name`,
       SUM(Sales) AS Total_Sales
FROM superstore
GROUP BY `Customer Name`
ORDER BY Total_Sales DESC
```

LIMIT 10;



The screenshot shows a 'Result Grid' interface with a 'Filter Rows' button. The grid displays a list of customers and their total sales, sorted in descending order. The first customer, Sean Miller, has the highest total sales at 24,205.62. The list includes 10 customers in total, with the last one partially visible as 'Sanjit Esale'.

	customer_name	Total_Sales
▶	Sean Miller	24205.62
	Tamara Chand	19017.85
	Raymond Buch	15117.35
	Tom Ashbrook	14588.58
	Ken Lonsdale	14049.92
	Sanjit Chand	13831.78
	Adrian Barton	13644.79
	Hunter Lopez	12873.30
	Sanjit Esale	12700.44

Brief Insights:

- Top-selling products contribute significantly to overall revenue.
- Technology/Furniture/Office Supplies categories can be ranked based on total sales.
- A small number of customers generate a large portion of sales.

6.Solve use cases (monthly trends, top customers, duplicates).

1. Monthly Sales Trend

```
SELECT MONTH(STR_TO_DATE(order_date, '%m/%d/%Y')) AS Month,  
        SUM(Sales) AS Total_Sales  
FROM superstore  
GROUP BY Month  
ORDER BY Month;
```

Result Grid		
	Month	Total_Sales
▶	1	77494.62
	2	54701.81
	3	187669.83
	4	130709.44
	5	140948.73
	6	127194.09
	7	137763.14
	8	139465.30
	9	272779.45

2. Top 10 Customers

```
SELECT customer_name,
SUM(Sales) AS Total_Sales
FROM superstore
GROUP BY customer_name
ORDER BY Total_Sales DESC
LIMIT 10;
```

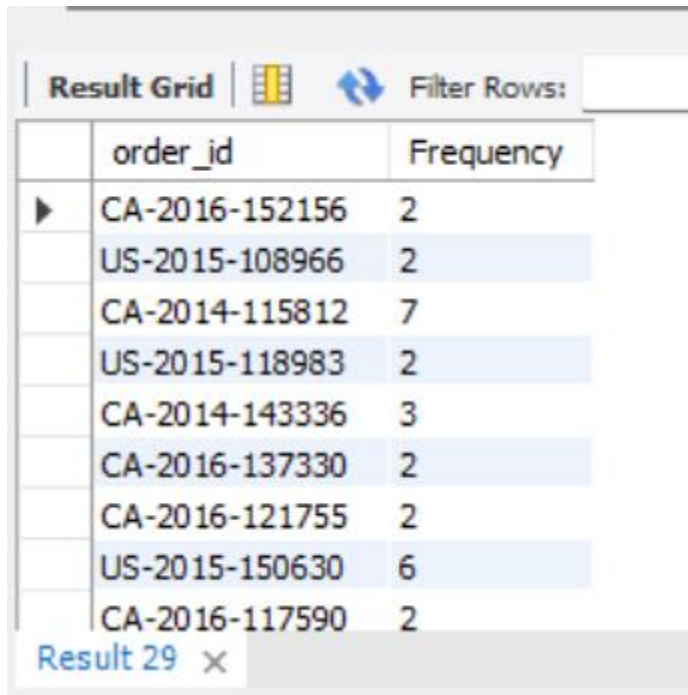
Result Grid		
	customer_name	Total_Sales
▶	Sean Miller	24205.62
	Tamara Chand	19017.85
	Raymond Buch	15117.35
	Tom Ashbrook	14588.58
	Ken Lonsdale	14049.92
	Sanjit Chand	13831.78
	Adrian Barton	13644.79
	Hunter Lopez	12873.30
	Saniit Endle	12209.44

3. Find Duplicate Orders

```

SELECT order_id,
       COUNT(*) AS Frequency
FROM superstore
GROUP BY order_id
HAVING COUNT(*) > 1;

```



The screenshot shows a 'Result Grid' window with a 'Filter Rows' button. The grid contains the following data:

	order_id	Frequency
▶	CA-2016-152156	2
	US-2015-108966	2
	CA-2014-115812	7
	US-2015-118983	2
	CA-2014-143336	3
	CA-2016-137330	2
	CA-2016-121755	2
	US-2015-150630	6
	CA-2016-117590	2

Result 29 x

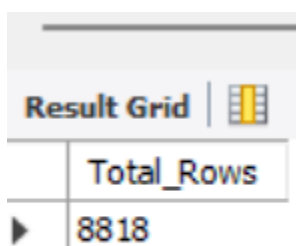
7. Validate results (row counts, data quality).

1. Total Number of Rows

```

SELECT COUNT(*) AS Total_Rows
FROM superstore;

```



The screenshot shows a 'Result Grid' window with the following data:

	Total_Rows
▶	8818

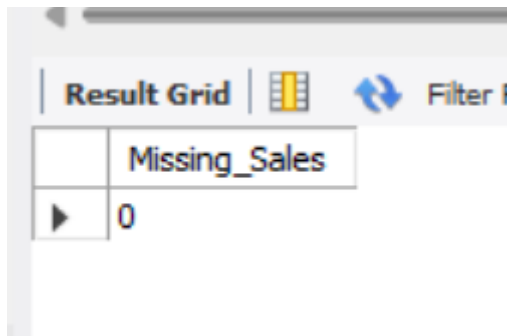
2. Check Missing Sales Values

```

SELECT COUNT(*) AS Missing_Sales
FROM superstore

```


WHERE Sales IS NULL;



A screenshot of a SQL query result grid. The grid has two columns: the first column is empty, and the second column is labeled 'Missing_Sales'. The first row of data shows a count of 0. Above the grid, there are tabs for 'Result Grid' and 'Filter', and a blue double-headed arrow icon.

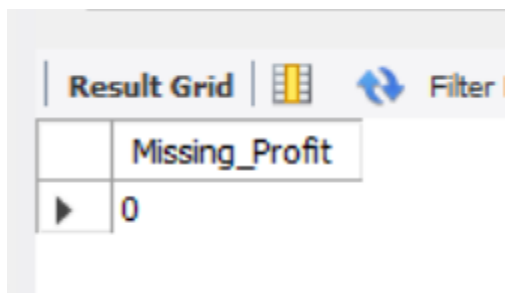
	Missing_Sales
▶	0

3. Check Missing Profit Values

```
SELECT COUNT(*) AS Missing_Profit
```

```
FROM superstore
```

```
WHERE Profit IS NULL;
```



A screenshot of a SQL query result grid. The grid has two columns: the first column is empty, and the second column is labeled 'Missing_Profit'. The first row of data shows a count of 0. Above the grid, there are tabs for 'Result Grid' and 'Filter', and a blue double-headed arrow icon.

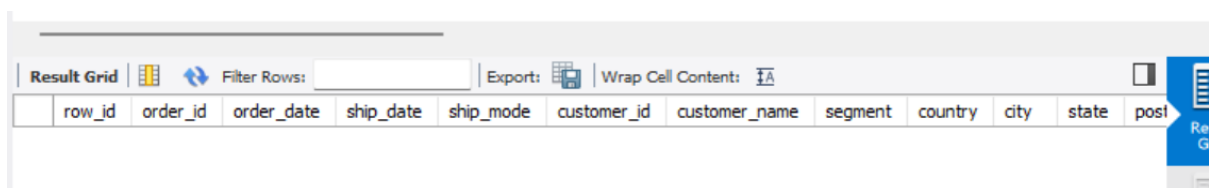
	Missing_Profit
▶	0

4. Check Negative Sales

```
SELECT *
```

```
FROM superstore
```

```
WHERE Sales < 0;
```



A screenshot of a SQL query result grid showing the columns of a dataset. The grid has a header row with the following column names: row_id, order_id, order_date, ship_date, ship_mode, customer_id, customer_name, segment, country, city, state, and post. Above the grid, there are tabs for 'Result Grid' and 'Filter', and a blue double-headed arrow icon. To the right of the grid, there are buttons for 'Export' and 'Wrap Cell Content'.

row_id	order_id	order_date	ship_date	ship_mode	customer_id	customer_name	segment	country	city	state	post
--------	----------	------------	-----------	-----------	-------------	---------------	---------	---------	------	-------	------

Brief Insights:

Monthly Trend

Sales vary across months, indicating seasonal demand patterns.

Top Customers

A small group of customers contributes a significant share of total sales.

Duplicates

Duplicate Order IDs were checked to ensure data consistency.